

Dwip Dalal

Ph.D. ECE @ UIUC, dwip2@illinois.edu

 GitHub  LinkedIn  Google Scholar  Website

Education

- **University of Illinois Urbana-Champaign (UIUC)** **2024 – Present**
Ph.D. Candidate, Electrical and Computer Engineering; Advisor: [Prof. Svetlana Lazebnik](#)
CGPA: 4.0/4.0
- **Indian Institute of Technology Gandhinagar (IIT-GN)** **2020 – 2024**
B.Tech in Mechanical Engineering with Minor in Computer Science
[Institute Gold Medalist](#); Overall CGPA: 9.1/10; Minor CGPA: 10/10

Interest Areas

Multimodal Representation Learning, VLAs, WAMs, VLMs, Reasoning, Test-Time Scaling

Awards & Achievements

- Awarded the *Dan Vivoli Endowed Fellowship* @ UIUC 2026–2027 for outstanding academic performance.
- Awarded of the *Qualcomm Graduate Award* @ UIUC 2025-2026.
- Awarded the *Dilip and Sandhya Sarwate Graduate Fellowship* @ UIUC 2024-2025, awarded to outstanding incoming graduate students.
- Awarded the scholarship from Thinking Machines to support my research on VLAs.
- Awarded the scholarship from Modal to support my research on VLAs.
- Awarded the [Institute Gold Medal](#) @ IIT Gandhinagar, for the highest CGPA in the discipline.
- Selected for Y Combinator's AI Startup School 2026.
- 1st position out of 200 participants in a ML hackathon organized by SmartSense Consulting Solutions Pvt. Ltd., earning a job offer for the role of ML Engineer II (declined).
- Institute nomination for Pre-Doctoral Research Assistant Program at Microsoft Research, India.
- 1st rank in [Undergraduate Research Showcase '23](#) @ IITGN for work on AI-enabled on-device drones.
- 10/10 GPA for two consecutive semesters (6th & 7th), each comprising rigorous 26-credit course work.
- Awarded *Shastri Indo-Canadian Institute Scholarship* for my research at the University of British Columbia.
- Travel Award from IIT Gandhinagar for attending ACL 2023, Toronto.
- 1st rank in [Hackrushi '22](#), the flagship annual machine learning hackathon of IIT Gandhinagar.
- 1st rank in IITGN Summer Project 2021 for building the most efficient NLP classification model.
- 1st rank in the competitive programming contest @ IIT-GN, participated by B.Tech, MTech, and Ph.D.

Research Internships

- **Applied Scientist Intern @ Amazon** (May 26 – Present)
Mentor: [Dr. Sowndarya Sundar](#) *Location:* Sunnyvale
Working in the Ring team, developing efficient video tokenization methods for MLLMs to improve long-video understanding by selectively preserving relevant visual tokens. Research paper in progress.
- **ML Research Intern @ Microsoft Research** (May 25 – Mar 26)
Mentor: [Dr. Nebojsa Jojic](#) (*Senior Principal Researcher*) *Location:* Redmond
Built CityNav, a benchmark for evaluating MLLMs on long-horizon outdoor navigation, and VoP, a method that elicits latent knowledge from MLLMs, achieving SOTA performance on instruction-free navigation [3].
- **ML Research Intern @ University of British Columbia (MITACS GRI)** (May 23 – Aug 23)
Mentor: [Prof. Yankai Cao](#) *Location:* Vancouver
Built a neural-to-video generation framework by learning the joint distribution between video and spike-train (brain activity) that reconstructs mouse behavior directly.
- **NLP Research Intern @ AI Institute, University of South Carolina** (Aug 22 – Feb 24)
Mentor: [Prof. Amit Sheth](#) (*Director of AIISC institute @ UoSC*), [Prof. Amitava Das](#), [Dr. Aman Chadha](#)
Worked on multimodal fact verification for language models, [7, 8]; multitask learning [14]. Also, launched [DeHate](#), a challenge focused on the automatic blurring of offensive segments (De-Factify 3.0 @AAAI'24).

Academic Research Positions

- **Research Assistant, University of Illinois Urbana-Champaign**

(Aug 24 – Present)

Mentor: Prof. Svetlana Lazebnik, Prof. Unnat Jain, Prof. Heng Ji

Improved Vision-Language-Action models for robotic Manipulation by aligning action, language, and vision representations [1]. Improved visual grounding in MLLMs through self-correction, guided by internal cross-attention [2]; improved compositional reasoning of MLLMs via joint image-language decomposition [4]; and authored a survey on agentic systems for automating data science workflows [9].

- **Research Assistant, IIT Gandhinagar**

(Nov 21 – May 24)

Mentor: Prof. Shanmuganathan Raman, Prof. Anirban Dasgupta, Prof. Mayank Singh, Prof. Harish PM

Worked on diffusion model for image-to-image generation tasks [6], reinforcement learning for control [13], self-supervised and cross-modal representation learning [5], normalizing flow for constrained geometry, density estimation and path-planning [10, 11], and multilingual topic modeling [12].

Publications

Under Review

1. Generalizable VLA Finetuning via Representation Anchoring and Language-Action Alignment

Dwip Dalal, S. Patel, C. Jain, J. Kim, U. Mishra, A. Baratian, H. Ha, Heng Ji, Svetlana Lazebnik, Unnat Jain

Under review

(Paper) (Project Page)

Conference Papers

2. Constructive Distortion: Improving MLLMs with Attention-Aware Image Warping

Dwip Dalal, G. Vashishtha, U. Mishra, J. Kim, M. Kanda, H. Ha, Svetlana Lazebnik, Heng Ji, Unnat Jain

ICLR 2026

(Paper) (Project Page)

3. City Navigation in the Wild: Exploring Emergent Navigation from Web-Scale Knowledge in MLLMs

Dwip Dalal*, Utkarsh Mishra*, Narendra Ahuja, Nebojsa Jojic

EACL 2026 (Oral)

(Paper) (Project Page)

4. Compositional Reasoning via Joint Image and Language Decomposition

Dwip Dalal*, Madhav Kanda*, Zhenhailong Wang, Heng Ji, Unnat Jain

EACL 2026

(Paper)

5. Learning Robust Deep Visual Representations from EEG Brain Recordings

Prajwal Singh, **Dwip Dalal**, Gautam Vashishtha, Shanmuganathan Raman, Krishna Prasad Miyapuram

WACV 2024

(Paper)

Featured in WACV Daily and Best of WACV 2024

6. Single Image LDR to HDR Conversion using Conditional Diffusion

Dwip Dalal, Gautam Vashishtha, Prajwal Singh, Shanmuganathan Raman

ICIP 2023 (Oral)

(Paper)

7. FACTIFY-5WQA: 5W Aspect-based Fact Verification through Question Answering

Anku Rani, SM Tonmoy, **Dwip Dalal**, Shreya Gautam, Megha C., Aman Chadha, Amit Sheth, Amitava Das

ACL 2023

(Paper) (Demo)

8. A Benchmark for Multimodal Fact Verification with Explainability through 5W Question-Answering

M Chakraborty, K Pahwa, A Rani, S Chatterjee, **Dwip Dalal**, ... A Chadha, Amit Sheth, Amitava Das

EMNLP 2023

(Paper)

Technical Reports

9. A Dataset-Centric Survey of LLM-Agents for Data Science

Dwip Dalal*, Chuxuan Hu*, Xiaona Zhou*

(Paper)

10. Flow Symmetrization for Parameterized Constrained Diffeomorphisms

Dwip Dalal*, Aalok Gangopadhyay*, Progyan Das*, Shanmuganathan Raman

(Paper) (Project Page)

Workshop Papers

11. ODESolvers are also Wayfinders: Neural ODEs for Multi-Agent Pathplanning

Dwip Dalal*, Progyan Das*, Anirban Dasgupta

NeurIPS 2023 Workshop (DLDE)

(Paper)

12. MMT: A Multilingual and Multi-Topic Indian Social Media Dataset
Dwip Dalal, Vivek Srivastava, Mayank Singh
EACL 2023 Workshop (C3NLP) [\(Paper\)](#)
13. Stabilizing Swirling Pendulum using Actor-Critic Method
Dwip Dalal, Shubhankar Riswadar, Harish J Palantheandalam-Madapusi
Indian Control Conference (ICC) 2023 [\(Paper\)](#)
14. SEPSIS: I can catch your lies - A new paradigm for Deception Detection
Anku Rani, **Dwip Dalal**, S. Gautam, P. Gupta, Vinija Jain, Aman Chadha, Amit Sheth, Amitava Das
ACL 2025 Workshop (SRW) [\(Paper\)](#)

Industrial Internships

- **ML Intern @ Physical Research Laboratory, ISRO** (Jan 23 – July 23)
Mentor: Prof. Dibyendu Chakrabarty (PI, ISRO Aditya-L1 Mission)
Implemented transformer-based pipeline with Bayesian inference on data from India's maiden solar probe, the Aditya-L1, to predict anomalous geomagnetic storms with uncertainty quantification.
- **ML Intern @ Tata Consultancy Services (TCS Research)** (Oct 22 – April 23)
Mentor: Dr. Manasi Patwardhan
Designed a transformer-based architecture for product copy generation, trained on the FACAD dataset. Fine-tuned BLOOM and ResNet models using the CLIP loss function to enable effective cross-modal learning.
- **AI Intern @ Defense Research and Development Organization (DRDO)** (Jan 22 – Aug 22)
Developed an RL framework using the actor-critic method in a PyBullet environment with simulated physics to learn drone swarm navigation.
- **SDE Intern @ Efcens Systems** (May 22 – Aug 22)
Developed an intelligent network threat detection platform to red-flag anomalies with Sanjeev Kumar, CTO of DELL, and Prasad Malempati, Sr. Manager of Accenture, San Francisco.

Relevant Courses

- **UIUC:** CS444 Deep Learning Computer Vision (**A+**); CS543 Computer Vision (**A+**); CS598 Topics in LLM Agents; CS546 Advanced Topics in NLP; CS 598 LLM Applications; ECE544 Pattern Recognition; ECE 598 Diffusion Model
- **IITGN:** Optimization for ML; Large Language Models; Digital Image Processing; Data Structures and Algorithms; Control Theory; Linear Algebra; Multivariable Calculus; Complex Analysis; Differential Equations; Probability; Numerical Methods

Open-Source Contributions

- **Merged Pull Request @ OpenVINO (Intel)** [\(Repository\)](#)
- **NeuralSight: Computer Vision Algorithm Library** [\(Repository\)](#)
- **JointGYM: Reinforcement Learning Algorithm Library** [\(Repository\)](#)
- **Jacobian Transpose Method for Controlling Robotic Manipulators** [\(Repository\)](#)

Social Initiatives, Leadership & Innovation

- **Instructor, SC 336: Mathematics of Machine Learning | IIT-GN | Fall, 2023** [\(certificate\)](#) [\(Page 25\)](#)
 - Instructed IIT Gandhinagar's first student-led, fully accredited short course, with class size of 170 students.
- **Founder & President of Society for Machine Learning, IIT-GN**
- **Reviewer:** CVPR'26, ICLR'26, EMNLP'25, NeurIPS'25, ICCV'25, ACL'25, AISTATS'25, ICML'25, NeurIPS'24
- **Teaching Assistant, UIUC:** CS 444: Deep Learning for CV (Fall'25), CS449: AI (Spring'25)
- **Teaching Assistant, IIT-GN:** ES413 Deep Learning (Spring'24), CS328 Data Science (Spring'23)
- **Core Member of Algorithms and Competitive Coding Club, IIT-GN** (Jun'22 - Mar'23)
- **Core Member of Robotics Club, IIT-GN** (Jun'22 - Mar'23)
- **Organizer of Hackrush'23 & '24, Annual Machine Learning Hackathon, IIT-GN**
- **Represented IIT-GN in the Inter-IIT Tech Meet in 2021, 2022, 2023 (Led team of 10).**